

AniTune: Robust Anime Face Recognition via Fine-Tuned Vision Transformer

Parameter-Efficient LoRA Adaptation on iCartoonFace

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1 Overview

We investigate the efficacy of Low-Rank Adaptation (LoRA) [1] for fine-tuning Vision Transformers (ViTs) [2] on the anime/cartoon face domain. Using *iCartoonFace*—389,678 images from 5,013 identities [7]—as our benchmark, we adapt a pretrained ViT backbone (e.g., MAE [3] or DINOv2 [4]) with parameter-efficient LoRA modules. Our goals are to (i) compare LoRA with full fine-tuning under constrained compute, and (ii) report closed-set identification and verification performance with transparent cost (trainable parameters, time, memory).

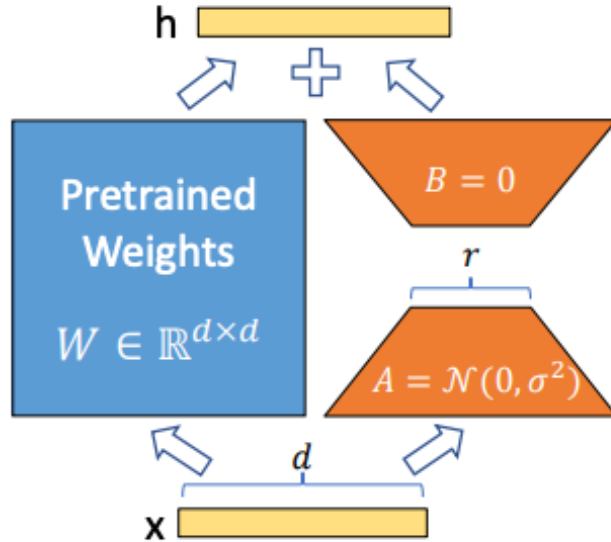


Figure 1: LoRA idea (*re-drawn/ adapted from [1]*). A frozen weight $W \in \mathbb{R}^{d \times d}$ is augmented with a trainable low-rank update BA (rank= r), where B is initialized to zero and A to $\mathcal{N}(0, \sigma^2)$.

2 Background

Cartoon/anime faces exhibit stylised color fields and geometry, large pose/occlusion variance, and a domain gap from natural images. The iCartoonFace benchmark [7] provides large-scale identities and annotations for recognition and detection. While CNNs with margin-based losses (e.g., ArcFace [5], CosFace [6]) remain strong in general face recognition, ViTs pretrained by MAE [3] or DINOv2 [4] show excellent transfer. However, parameter-efficient adaptation in stylised domains is less explored; we target this gap with LoRA on ViTs.

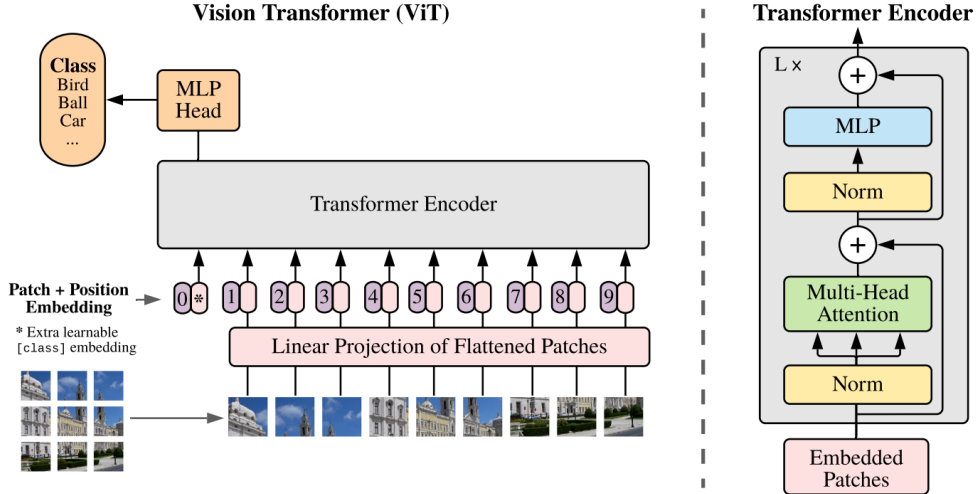


Figure 2: ViT architecture (adapted from [2]). Images are split into patches, linearly projected, summed with positional embeddings, and processed by a Transformer encoder; a head predicts class labels.

3. Statement of Work

3.1 Dataset

We use *iCartoonFace* [7] (389,678 images, 5,013 identities). We adopt its recognition split, clean faces by center-crop/ alignment, apply light augmentations, and ensure disjoint IDs for open-set testing. The distribution is long-tailed (many IDs < 30 images).

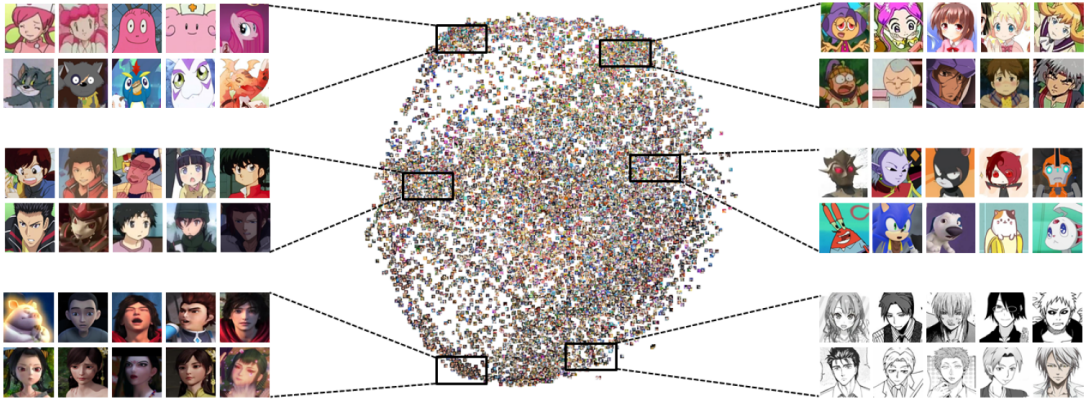


Figure 3: iCartoonFace t-SNE montage (reproduced/ adapted from [7]).

3.2 Method

Prior work: Recognition commonly uses CNN backbones with margin-based softmax [5, 6]. The original CartoonFace benchmark used domain-adaptive methods [7].

Planned work:

- Select a pretrained ViT backbone (MAE ViT-B/16 [3] or DINOv2 ViT-B/14 [4]).
- Insert LoRA in attention (qkv/proj) with rank $r \in \{4, 8, 16\}$ [1].
- Freeze backbone; train LoRA modules + recognition head on iCartoonFace.

- Compare against a small-scale full fine-tuning baseline.
- Evaluate Top-1/Top-5, macro-F1, and TAR@FAR (1%). Log trainable params, GPU memory, and time.



Figure 4: Animate face detection/recognition outcomes. If from prior work or toolkit, add the correct citation here (e.g., dataset/tool paper). If this collage is *yours*, replace with “(ours)” and remove the citation.

3.3 Outcome & Evaluation

We expect LoRA to recover a large fraction of full fine-tuning accuracy with far fewer trainable parameters and lower cost. Success target: achieve $\geq 90\%$ of full-FT Top-1 with $\leq 10\%$ trainable params and sizable savings in wall-clock time/VRAM.

4. Project Plan

Week	Task
W1	Setup GitHub repo, code base (ViT+LoRA); download/explore dataset; implement baseline: freeze backbone, train head on ViT + iCartoonFace.
W2	Insert LoRA ($r \in \{4, 8, 16\}$) into ViT; run adaptation; log performance and parameters/time.
W3	Full fine-tuning baseline (backbone+head) at smaller scale; compare to LoRA; ablations: LoRA rank, input resolution.
W4	Aggregate results; plots (accuracy vs parameters/time); finalize report; release code/configs.

GitHub URL: <https://github.com/Ryan-GRY/AniTune.git>

Task	Week 1	Week 2	Week 3	Week 4
Setup GitHub & Env.	100%			
Baseline Head-Only	50%			
LoRA Impl. & Experiments		0%		
Full Fine-Tuning			0%	
Ablations & Latency			0%	
Aggregate Results & Write-up				0%

Table 1: Milestone progress grid (column widths fixed to avoid overflow).

6. Contributions

- **Tingting Du:** LoRA modules in ViT; dataset preprocessing; baseline head-only training.
- **Evelyn Liu:** LoRA rank/resize ablations; latency/throughput logging.
- **Ryan Gao:** Full fine-tuning baseline; ablations; final write-up and release.

References

- [1] Edward J. Hu, Yelong Shen, et al. *LoRA: Low-Rank Adaptation of Large Language Models*. ICLR, 2022 (original preprint 2021).
- [2] Alexey Dosovitskiy, et al. *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. ICLR, 2021.
- [3] Kaiming He, Xinlei Chen, et al. *Masked Autoencoders Are Scalable Vision Learners*. CVPR, 2022.
- [4] Maxime Oquab, et al. *DINOv2: Learning Robust Visual Features without Supervision*. TMLR, 2023.
- [5] Jiankang Deng, et al. *ArcFace: Additive Angular Margin Loss for Deep Face Recognition*. CVPR, 2019.
- [6] Hao Wang, et al. *CosFace: Large Margin Cosine Loss for Deep Face Recognition*. CVPR, 2018.
- [7] Y. Zheng, Y. Zhao, et al. *Cartoon Face Recognition: A Benchmark Dataset*. arXiv:1907.13394v3, 2020.